

EXPLORING DATA VIA INVARIANT COORDINATE SELECTION IN R, PYTHON, AND JULIA

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joint w/ Anne M. Ruiz

ICS and Related Methods Conference
May 28, 2026



OUTLINE OF THE PRESENTATION

Invariant coordinate selection

- Motivation

- General principle

- Illustrations

- ICS - methodology

- Applications

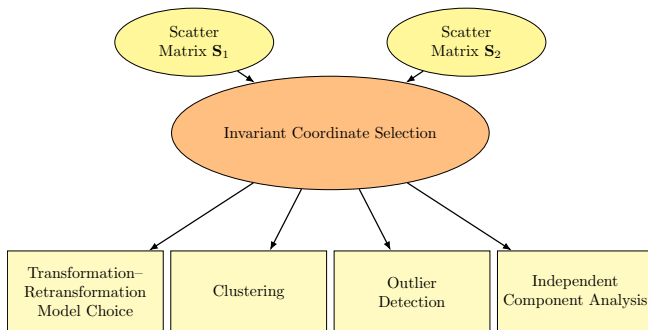
ICS - implementations

Conclusion

Tutorial

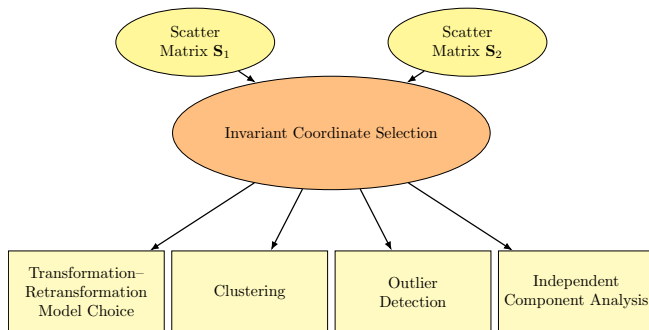
INVARIANT COMPONENT SELECTION (ICS)

ICS Tyler et al. (2009) - GPCA Caussinus and Ruiz-Gazen (1990)



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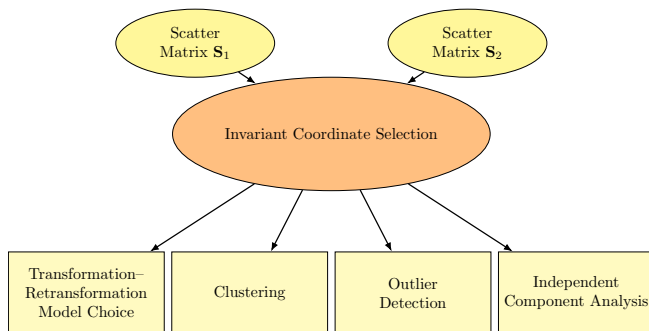
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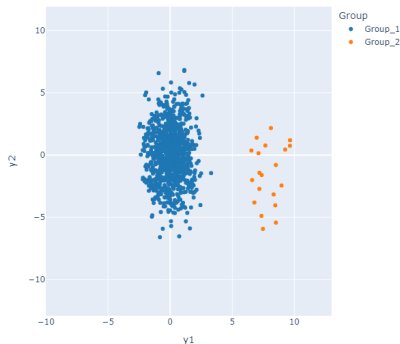
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- **Affine invariant** multivariate method.
- Can recover the **Fisher's linear discriminant subspace** under some elliptical mixture models, without knowing the class labels (Tyler et al., 2009; Becquart et al., 2024).

EXAMPLE WITH 2 GROUPS - 2 DIMENSIONS

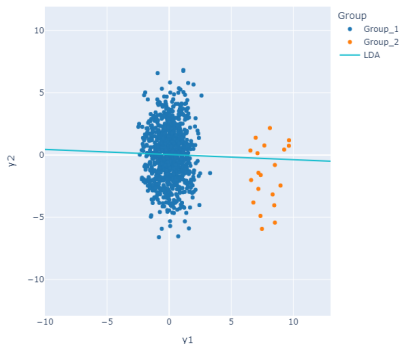


Linear Discriminant Analysis

- ▶ **Supervised** learning
- ▶ Finds a lower-dimensional subspace that maximizes the separation between different classes (FDS)
- ▶ Maximizes the ratio:

$$\arg \max_{v \in \mathbb{R}^p} \frac{v' B v}{v' W v}$$

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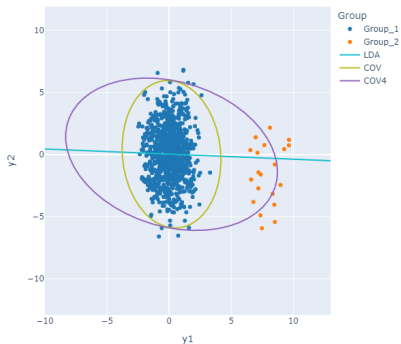


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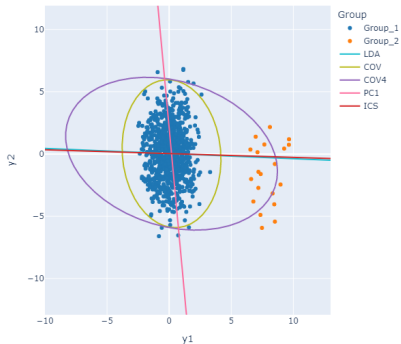
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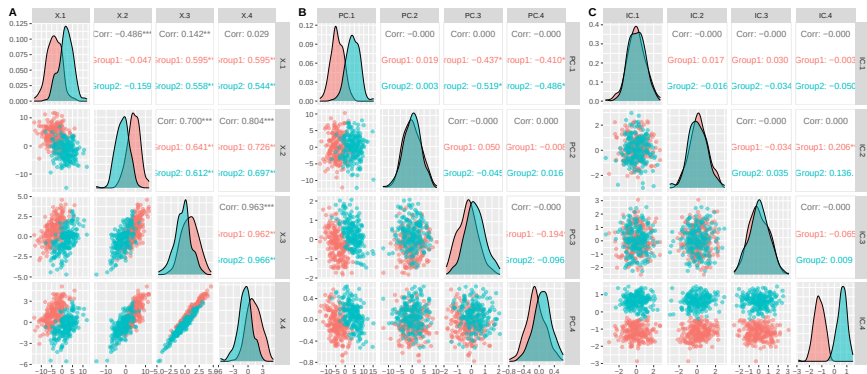
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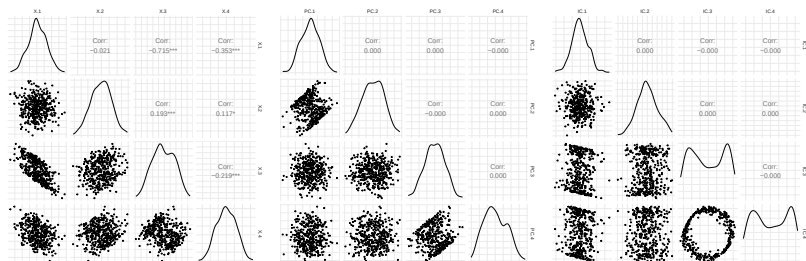
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TOY EXAMPLE FROM NORDHAUSEN AND RUIZ-GAZEN (2021)



ANOTHER TOY EXAMPLE



1. Components calculation

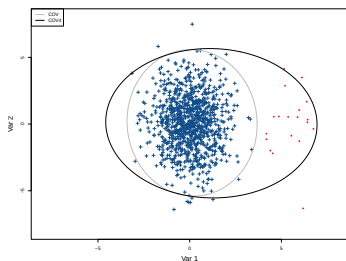
1. Two scatter matrices $\mathbf{V}_1$ and $\mathbf{V}_2$.
2. Eigendecomposition of $\mathbf{V}_1^{-1}\mathbf{V}_2$.
3. Decreasing eigenvalues: kurtosis.
4. Projection of the centered data on the eigenvectors $\rightarrow$ ICs.

1. Components calculation

1. V_1 : covariance matrix.
2. V_2 : fourth moment matrix.
3. Eigendecomposition of $V_1^{-1}V_2$
4. Projection of the centered data on the eigenvectors.

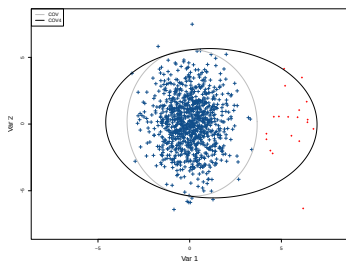
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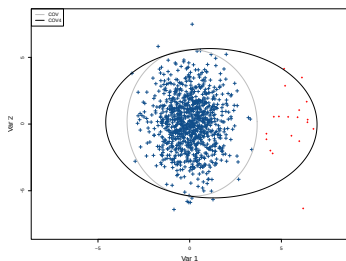


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3. **First and/or last $k \leq p$ components retained.**

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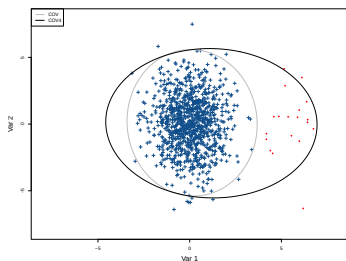


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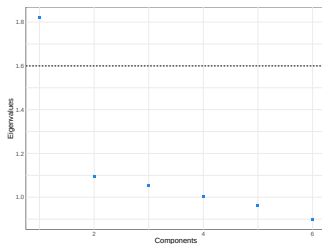
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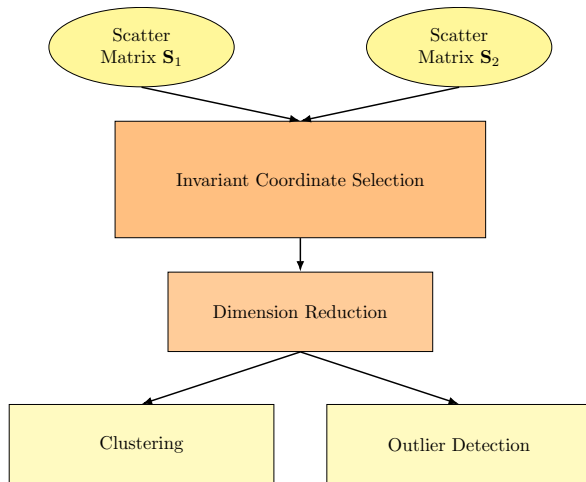
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→ **Depends on the application.**

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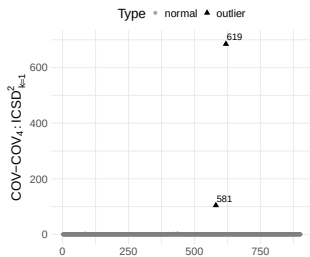
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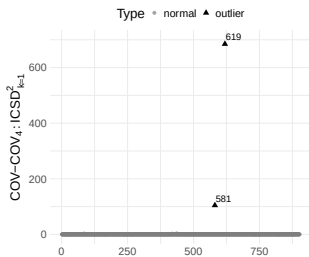
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Tandem clustering with ICS

1. Dimension reduction to retain k selected ICs.
2. Clustering method.
3. Identification of clusters.

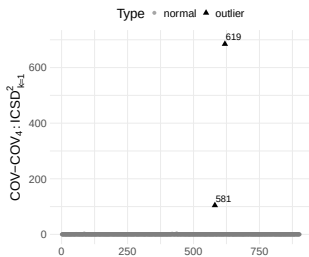
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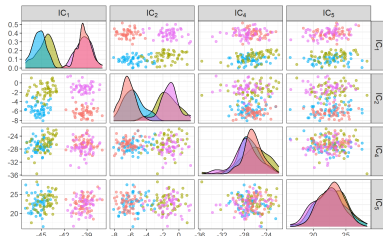
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Ecosystem

Software design

R

Python

Julia

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ICS ECOSYSTEM DEVELOPMENT



SOFTWARE DESIGN

Main common features:

- ▶ the `ICS` class;
- ▶ several scatter matrices;
- ▶ supported algorithms: `"whiten"`, `"standard"`, `"QR"`;
- ▶ same parameters: `fix_signs`, `center`.

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Main differences:

- ▶ **R**: coexistence of S3 and S4 class system
- ▶ **Python**: computing the invariant coordinates is divided into `fit` and `transform` methods, in line with popular machine learning frameworks such as scikit-learn:
- ▶ **Julia**: same approach as Python with a `ICSModel` class.

ICS - ALGORITHMS

Let $\mathbf{X}_n = (\mathbf{x}_1, \dots, \mathbf{x}_n)'$ be a p -variate dataset.

Simultaneous diagonalization of two scatter matrices $\mathbf{S}_{1,n}$ and $\mathbf{S}_{2,n}$:

$$\mathbf{W}_n \mathbf{S}_{1,n} \mathbf{W}_n^\top = \mathbf{I}_p \quad \text{and} \quad \mathbf{W}_n \mathbf{S}_{2,n} \mathbf{W}_n^\top = \mathbf{D}_n$$

where the diagonal matrix $\mathbf{D}_n$ contains the eigenvalues $\rho_1, \dots, \rho_p$ of $\mathbf{S}_{1,n}^{-1} \mathbf{S}_{2,n}$ in decreasing order and $\mathbf{W}_n = (\mathbf{w}_1, \dots, \mathbf{w}_p)'$ contains the corresponding eigenvectors as its rows.

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The (affine) invariant coordinates or scores are:

$$\mathbf{Z}_n = \mathbf{X}_n^c \mathbf{W}_n^\top.$$

$\mathbf{X}_n^c$ can be the centered version of $\mathbf{X}_n$ with respect to the location estimator associated with $\mathbf{S}_{1,n}$.

ICS - ALGORITHMS

- ▶ `whiten`: whitens X_n with respect to $\mathbf{S}_1$ before computing $\mathbf{S}_2$ (should be a function).
- ▶ `standard`: performs the spectral decomposition of the symmetric matrix $M(X_n)$.

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whiten:

- ▶ $Y_n = X_n S_1(X_n)^{-1/2}$
- ▶ $S_2(Y_n)$
- ▶ $S_2(Y_n) = UDU'$
- ▶ $W = U' S_1(X_n)^{-1/2}$
- ▶ $Z = X_n^c W^\top$

standard:

- ▶ $S_1(X_n), S_2(X_n)$
- ▶ $M(X_n) = S_1(X_n)^{-1/2} S_2(X_n) S_1(X_n)^{-1/2}$
- ▶ $M(X_n) = UDU'$
- ▶ $W = U' S_1(X_n)^{-1/2}$
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ICS - ALGORITHMS

The signs of W can be fixed with `fix_signs`:

- ▶ **"scores"**: the generalized skewness values of all components are positive.
Common for **ICS** framework.
- ▶ **"W"**: the maximum element in each row of W is positive and each row has norm 1.
Common with **ICA** framework.

Z can be centered through the boolean `center` input parameter.

ICS - **QR** - BASED ON PIVOTED QR DECOMPOSITION

Archimbaud et al. (2023b), numerically stable algorithm for $\text{COV} - \text{COV}_w$ focusing on:

$$M(X_n) = \text{COV}^{-1/2} \text{COV}_w \text{COV}^{-1/2}.$$

ICS - **QR** - BASED ON PIVOTED QR DECOMPOSITION

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Algorithm:

- ▶ Pivoted QR factorization: $\Pi_2^\top \left(\frac{1}{\sqrt{n-1}} X_n^c \right) \Pi_1 = QR$
- ▶ $Q = \Pi_2 Q, R = R \Pi_1^\top$
- ▶ Leverage scores $q_i = \|Q(i, :)\|_2^2, i = 1, \dots, n$
- ▶ SVD of $\text{Diag} \left(\sqrt{w((n-1)q_i)} \right)_{i=1}^n Q$ to obtain

$$\tilde{M}(X_n) = \frac{n-1}{n} Q^\top \text{Diag}(w((n-1)q_i))_{i=1}^n Q = \tilde{U}_2 \tilde{D}_2 \tilde{U}_2^\top$$

- ▶ $W = (R^{-1} \tilde{U}_2)^\top$
- ▶ $Z = \sqrt{n-1} Q \tilde{U}_2$ or equivalently $Z = X_n^c \Pi_1 W^\top$.

ICS - SUMMARY

Main attributes

Language Package	R ICS - <code>ics()</code> , <code>ics2()</code>	R ICS - <code>ICS()</code>	Python ICSpyLab	Julia ICSTools.jl
Class	'ICS'	'ICS'	'ICS'	'ICSModel'
Generalized eigenvalues	@gKurt	\$gen.kurtosis, gen.kurtosis()	kurtosis_	kurtosis_
Unmixing ma- trix	@UnMix	\$W	<code>components_</code>	W_
Scores	@Scores	\$scores, components()		scores_

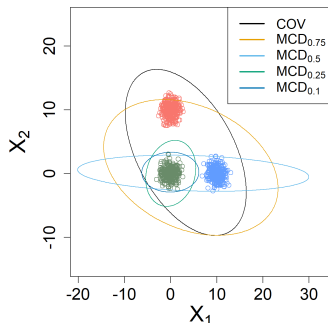
ICS - SUMMARY

Methods

Language Package	R ICS	Python ICSpyLab	Julia ICSTools.jl
Print basic information	<code>print()</code>		
Summary	<code>summary()</code>	<code>describe()</code>	<code>show()</code>
Coefficient matrix of ICS	<code>coef()</code>		
Plots the kurtosis measures	<code>screeplot()</code>	<code>plot_kurtosis()</code>	<code>scree_plot()</code>
Scatterplot of ICs	<code>plot()</code>		
Fit the ICS model		<code>fit()</code>	<code>fit!()</code>
Transform data using the fitted transformation matrix		<code>transform()</code>	<code>predict!()</code>
Fit and transform		<code>fit_transform()</code>	<code>fit_predict!()</code>
Extract the transformed data matrix	<code>fitted()</code>		
Transform data back to its original space		<code>inverse_transform()</code>	

CHOICE OF SCATTER MATRICES

- ▶ $\text{COV} - \text{COV}_4$ or FOBI Peña et al. (2010); Cardoso (1989); Nordhausen and Virta (2019); Fischer et al. (2017a, 2020)
- ▶ $\text{MCD}_\alpha - \text{COV}$, $\text{RMCD}_\alpha - \text{COV}$ with values of $\alpha \leq 0.5$ Rousseeuw (1985)
- ▶ $\text{TCOV} - \text{UCOV}$, $\text{TCOV} - \text{COV}$ Caussinus and Ruiz-Gazen (1993a)
- ▶ $\text{LCOV} - \text{COV}$ Hennig (2009)



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Applications:

- For **outlier detection**: Archimbaud et al. (2018a) for a small proportion of outliers.

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Applications:

- ▶ For **outlier detection**: Archimbaud et al. (2018a) for a small proportion of outliers.
- ▶ For **clustering**: Becquart et al. (2024) and Alfons et al. (2024). One scatter matrix should reflect the **within (local) cluster structure** and one the **between (global) cluster structure**.

CLASS AND RELEVANT COMPONENTS FOR SCATTER MATRICES

Language Package	R ICS	Python ICSpyLab	Julia ICSTools.jl	
Class	'ICS_scatter'	'Scatter'	'Scatter'	
Components	location	location	location	Estimated location vector (optional)
	scatter	scatter	scatter	Estimated scatter matrix
	label	label	label	Label for the scatter matrix (used for printing output)

Language Class		R	R	Python	Julia
			'ICS_scatter'	'Scatter'	'Scatter'
Covariance		ICS::cov()	ICS::ICS_cov()	cov()	cov2()
Fourth-moment	covariance	ICS::cov4()	ICS:: ICS_cov4()	cov4()	cov4()
One-step M-estimator		ICS:: covW()	ICS:: ICS_covW()	covW()	covW()
One-step Tyler shape matrix		ICS:: covAxis()	ICS:: ICS_covAxis()	covAxis()	covAxis()
Cauchy location and scatter		ICS::tM()	ICSClust:: ICS_mlc()		mlc()
Multivariate t-distribution estimator		ICS:: tM()	ICS:: ICS_tM()	tM()	tM()
Minimum Covariance Determinant		rrcov:: CovMed()	ICSClust:: ICS_mcd_raw(), ICS_mcd_rwt()	mcd()	mcd_raw(), mcd_rwt()
Pairwise one-step M-estimate		ICSClust:: tcov()	ICSClust:: ICS_tcov()	tcov()	tcov()
Simple robust estimates		ICSClust:: ucov()	ICSClust:: ICS_ucov()		
Local shape scatter		ICSClust:: lcov()	ICSClust:: ICS_lcov()		lcov()
Supervised scatter (quantiles)		ICS:: scovq()	ICS:: ICS_scovq()		

CHOICE OF COMPONENTS WITH $q < p$

Archimbaud et al. (2018a), Alfons et al. (2024), Nordhausen et al. (2022), Radojicic and Nordhausen (2020)

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Based on the Invariant Components:

- Keep only non-gaussian components using marginal normality tests.

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Based on the Invariant Components:

- Keep only non-gaussian components using marginal normality tests.

In this context of particular sequential multiple testing, we apply the Bonferroni correction on the significance level: $\alpha_i = \alpha/i$ for $i = 1, \dots, p$ with $\alpha = 5\%$ (see Dray (2008)).

CHOICE OF COMPONENTS WITH $q < p$

Based on the eigenvalues and eigenvectors:

- ▶ Visually, using a **scree plot**.
- ▶ Using the number q of clusters a priori and the fact that some should be almost equal (**med_criterion** and **var_criterion**).
- ▶ Using asymptotic distribution of the eigenvalues or bootstrapping (**ICSboot**, **FOBIasymp**, **FOBIboot**).
- ▶ Using quasi inferential procedures (**parallel analysis**).
- ▶ Using ladle and data augmentation (**FOBIladle**).

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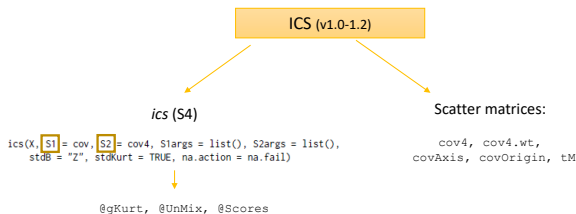
For clustering:

- ▶ **First** and/or **last** components

SELECTION OF COMPONENTS - SUMMARY IN R

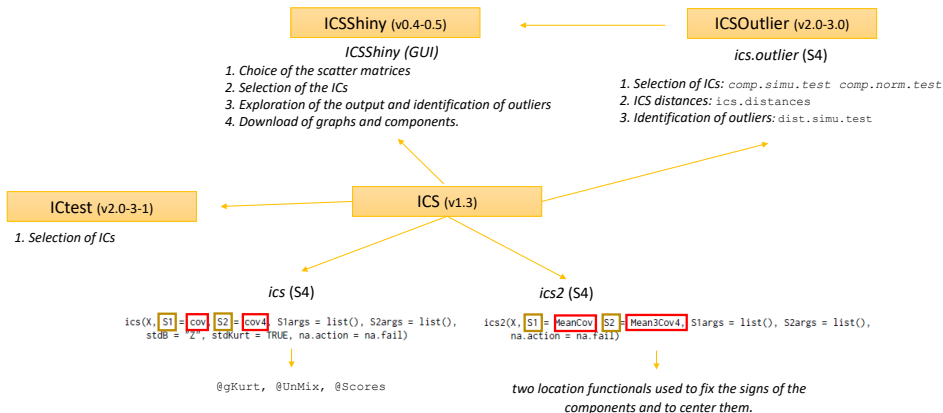
Criterion		Function	Notes
Marginal tests	normality	<code>ICSClust::normal_crit()</code>	First and last few components are investigated
Median-based criterion		<code>ICSClust::med_crit()</code>	Requires a priori knowledge
Variance-based criterion		<code>ICSClust::var_crit()</code>	From <code>ICtest::ICSboot()</code>
Discriminatory power criterion		<code>ICSClust::discriminatory_crit()</code>	Supervised, requires a priori knowledge
Parallel analysis (simulation)		<code>ICSOutlier::comp_simu_test()</code> , <code>comp.simu.test()</code>	Only first components are investigated
Marginal tests	normality	<code>ICSOutlier::comp_norm_test()</code> , <code>comp.norm.test()</code>	Only first components are investigated
Asymptotic and resampling based approaches		<code>ICtest::ICSboot()</code> , <code>FOBIboot()</code> , <code>FOBIasympt()</code> , <code>FOBIladle()</code>	Focus on identifying Gaussian subspace

GENESIS: 2007/2008 - KLAUS NORDHAUSEN + HANNU OJA + DAVID E. TYLER



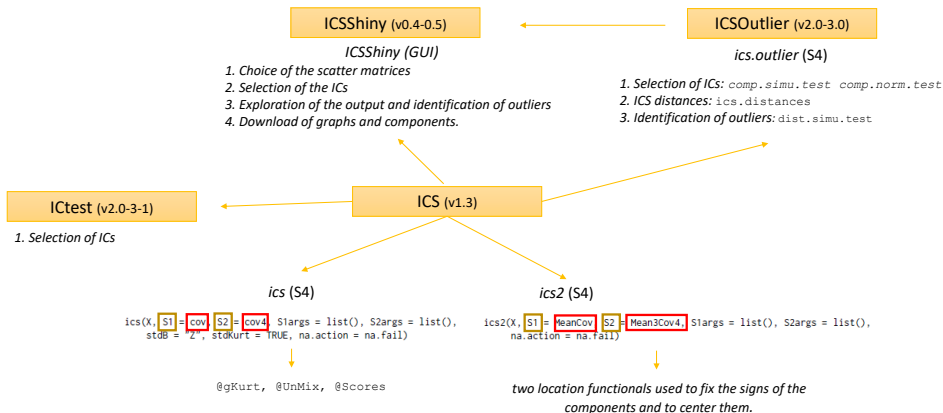
2016-2018: + AURORE ARCHIMBAUD AND ANNE RUIZ-GAZEN

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+ Joris May



2023: + ANDREAS ALFONS + ZLATKO DRMAČ

ICSClust (v1.0)

ICSClust(S3)

ICS_scatter: ICS_lcov, ICS_med, ICS_mlc, ICS_teov, ICS_ucov

1. *Computation of ICS*: ICS

2. *Selection of ICS and visualization*:

"med_crit", "normal_crit", "var_crit", "discriminatory_crit"

3. *Clustering and visualization*:

"kmeans_clust", "tkmeans_clust", "pam_clust", "mclust_clust",
"rmclust_clust", "rimle_clust"

ICSOutlier (v4.0)

ICS_outlier(S3)

0. *Computation of ICS*: ICS

1. *Selection of ICS*: comp_simu_test comp_norm_test

2. *ICS distances*: ics_distances

3. *Identification of outliers*: dist_simu_test

ICS (v1.4-2)

```
ICS(
  X,
  S1 = ICS_cov,
  S2 = ICS_cov4,
  S1_args = list(),
  S2_args = list(),
  algorithm = c("whiten", "standard", "QR"),
  center = FALSE,
  fix_signs = c("scores", "W"),
  na.action = na.fail
)
```

```
@gKurt   → $gen_kurtosis
@UnMix   → $W
@Scores  → $scores
```

ICS (S3)

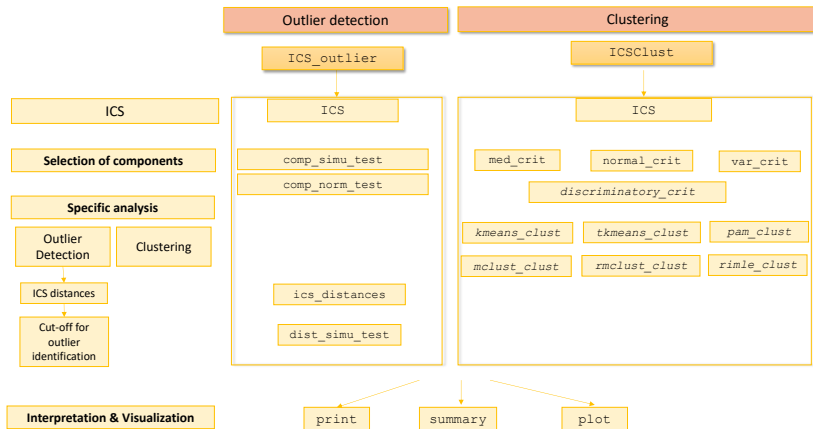
1. *ICS_scatter* class

2. *Different algorithms*

3. *Centering*

4. *Fixing signs*

ICS - R ECOSYSTEM



PYTHON - 2023: COLOMBE BECQUART + ABDALLAH ABDELSAMEIA

Package **ICSpyLab**

- ▶ version 0.1.0: May 2025
- ▶ version 1.0.0: May 2026
- ▶ Main features:
 - ▶ the ICS class;
 - ▶ several scatter matrices;
 - ▶ supported algorithms: "whiten", "standard", "QR" and "eigh".
 - ▶ same parameters: fix_signs, center.



JULIA 2026: VALENTIN TODOROV

Package **ICSTools.jl**

- ▶ Main features:
 - ▶ the `ICSTools` class;
 - ▶ several scatter matrices (**Robustbase.jl**);
 - ▶ supported algorithms: "whiten", "standard", "QR";
 - ▶ same parameters: `fix_signs`, `center`.



OUTLINE OF THE PRESENTATION

Invariant coordinate selection

ICS - implementations

Conclusion

Tutorial

CONCLUSION

ICS is an attractive **unsupervised** multivariate method:

- ▶ Designed to **find structure** in a low-dimensional subspace:
 - ▶ for outlier detection,
 - ▶ as pre-processing for clustering
- ▶ Affine invariant.

CONCLUSION

ICS is an attractive **unsupervised multivariate method:**

- ▶ Designed to **find structure** in a low-dimensional subspace:
 - ▶ for outlier detection,
 - ▶ as pre-processing for clustering
- ▶ Affine invariant.

Still presents some challenges:

- ▶ Choice of scatter matrices.
- ▶ Choice of components.

SOME PERSPECTIVES

Implementation:

- ▶ **R ecosystem**: algorithms for functional, compositional, collinear or HDLSS data.
- ▶ **Python** and **Julia**: utilities for outlier detection and clustering

SOME PERSPECTIVES

Implementation:

- ▶ **R ecosystem**: algorithms for functional, compositional, collinear or HDLSS data.
- ▶ **Python** and **Julia**: utilities for outlier detection and clustering

Github repository:

- ▶ <https://github.com/AuroreAA/ICS-implementation>

Working paper:

- ▶ Alfons, A., Archimbaud, A., Becquart, C., Nordhausen, K., & Todorov, V. (2026). *Exploring data via invariant coordinate selection in R, Python, and Julia*.

All contributions are welcomed.

OUTLINE OF THE PRESENTATION

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NOTATIONS

- ▶ $\mathbf{X}$ is a p -dimensional random vector, $F_{\mathbf{X}}$ its cumulative distribution function and $\mathbf{m}(F_{\mathbf{X}})$ an affine equivariant location estimator.
- ▶ $\mathbf{X}_n = (\mathbf{x}_1, \dots, \mathbf{x}_n)'$ a p -variate dataset. $\mathbf{x}_1, \dots, \mathbf{x}_n$ follow the same distribution as $\mathbf{X}$.

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- ▶ $\mathcal{P}_p$ is the set of all symmetric positive definite matrices of order p .
- ▶ A **scatter functional** is defined as a matrix $\mathbf{S}(F_{\mathbf{X}}) \in \mathcal{P}_p$, uniquely defined at $F_{\mathbf{X}}$, which is **affine equivariant** in the sense that:

$$\mathbf{S}(F_{\mathbf{A}\mathbf{X}+\boldsymbol{\gamma}}) = \mathbf{A}\mathbf{S}(F_{\mathbf{X}})\mathbf{A}',$$

for all $p \times p$ non-singular matrices $\mathbf{A}$ and all $\boldsymbol{\gamma} \in \mathbb{R}^p$.

For sake of simplicity, the dependence on $F_{\mathbf{X}}$ is dropped.

CHOICE OF SCATTER MATRICES I

- $\text{COV} - \text{COV}_4$ or FOBI Peña et al. (2010); Cardoso (1989); Nordhausen and Virta (2019); Fischer et al. (2017a, 2020)

$$\text{COV}_4(\mathbf{X}_n) = \frac{1}{n} \sum_{i=1}^n r^2(\mathbf{x}_i)(\mathbf{x}_i - \bar{\mathbf{x}}_n)(\mathbf{x}_i - \bar{\mathbf{x}}_n)^\top,$$

where $r^2(\mathbf{x}_i) = (\mathbf{x}_i - \bar{\mathbf{x}}_n)^\top \text{COV}(\mathbf{x}_n)^{-1}(\mathbf{x}_i - \bar{\mathbf{x}}_n)$ is the squared Mahalanobis distance.

CHOICE OF SCATTER MATRICES II

- $\text{MCD}_\alpha - \text{COV}$ and $\text{RMCD}_\alpha - \text{COV}$: (Reweighted) Minimum Covariance Determinant **Rousseeuw (1985)**.

$$\text{MCD}_\alpha(\mathbf{X}_n) = c_\alpha \frac{1}{n_\alpha} \sum_{j=1}^{n_\alpha} (\mathbf{x}_{i_j} - \bar{\mathbf{x}}_{\alpha,n})(\mathbf{x}_{i_j} - \bar{\mathbf{x}}_{\alpha,n})^\top,$$

where $n_\alpha = \lceil \alpha n \rceil$ observations for which the sample covariance matrix has the smallest determinant and usually $\alpha \in [0.5, 1]$.

In our ecosystem it is also interesting to consider **values of α that are smaller than 0.5**.

CHOICE OF SCATTER MATRICES III

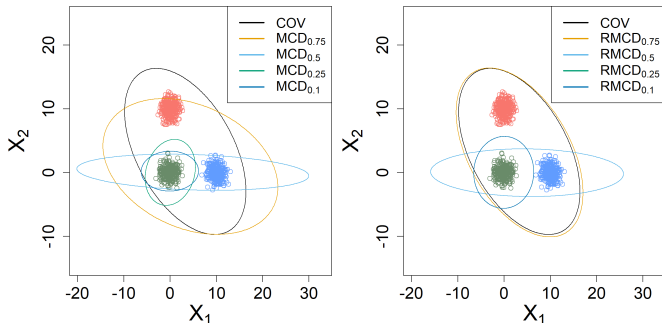


Figure: Shapes of different scatter matrices for a sample from a bivariate Gaussian mixture distribution with three balanced clusters.

CHOICE OF SCATTER MATRICES IV

- $SCOV - COV$, $TCOV - UCOV$ Caussinus and Ruiz-Gazen (1993a, 1995, 2007); Ruiz-Gazen (1996); Caussinus and Ruiz-Gazen (1990); Fekri and Ruiz-Gazen (2015)

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$$\text{TCOV}_{\beta}(\mathbf{X}_n) = \frac{\sum_{i=1}^{n-1} \sum_{j=i+1}^n w(\beta r^2(\mathbf{x}_i, \mathbf{x}_j)) (\mathbf{x}_i - \mathbf{x}_j)(\mathbf{x}_i - \mathbf{x}_j)^{\top}}{\sum_{i=1}^{n-1} \sum_{j=i+1}^n w(\beta, r^2(\mathbf{x}_i, \mathbf{x}_j))},$$

where $r^2(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i - \mathbf{x}_j)^{\top} \text{COV}(\mathbf{x}_n)^{-1} (\mathbf{x}_i - \mathbf{x}_j)$,
 $w(x) = \exp(-x/2)$, $\beta = 4$.

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 $w(x) = \exp(-x/2)$, $\beta = 4$.

- **LCOV – COV Hennig (2009)**, a local shape matrix based on the aggregation of covariance matrices computed based on 10% of the nearest neighbors (with regard to the Mahalanobis distance).

PARALLEL ANALYSIS (PA)

As in [Peres-Neto et al. \(2005\)](#) for PCA or in [Caussinus et al. \(2003\)](#).

Computation of cut-offs:

- 10 000 simulations of $\mathcal{N}_n(0, I_p)$
- ICS with $\text{COV}(\mathbf{X}_n)^{-1} \text{COV}_4(\mathbf{X}_n)$
- Quantiles of the ICS eigenvalues at level $1 - \frac{\alpha}{i}$ for each component i , with $\alpha = 5\%$ and $i = 1, \dots, p$.

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Test:

- Sequentially testing if the ICS eigenvalues are higher than corresponding quantiles.
- Stop as soon as one is lower than the cut-off.

NORMALITY TESTS

Finding the first i^{th} coordinate with no longer information about the structure of the data through normality tests.

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Normality tests:

- The **D'Agostino** test of skewness (**DA**),
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- The **Bonett-Seier** (**BS**) test of Geary's kurtosis,
- The **Jarque-Bera** (**JB**) test for normality which is based on both skewness and kurtosis measures,
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Stopping rule

- Sequentially testing if the ICs are gaussian or not.
- Stop as soon as one is gaussian based on the corrected level of 5%.